



Effect of Al addition on mechanical properties and microstructure of refractory $\text{Al}_x\text{HfNbTaTiZr}$ alloys



Chun-Ming Lin*, Chien-Chang Juan, Chia-Hsiu Chang, Che-Wei Tsai, Jien-Wei Yeh*

Department of Materials Science and Engineering, National Tsing Hua University, Hsinchu 30013, Taiwan

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ABSTRACT

This study developed a series of refractory $\text{Al}_x\text{HfNbTaTiZr}$ high-entropy alloys (HEAs) with an aim to improve strength, and reduce density of the very ductile base alloy HfNbTaTiZr . Despite the diversity of crystal structures among the constituent elements, all the HEAs are single solid solution phase with body-centered cubic (BCC) structure. The addition of Al significantly improves the strength but reduces the ductility due to large solution hardening. The linear relation between yield strength and atomic percentage of Al suggests that the strengthening effect of a certain element in a single-phase HEA alloy can be explained based on quasi-binary alloy concept. Crack formations in deformed AlHfNbTaTiZr alloy with the lowest fracture strain are mainly along the boundaries between dendrite and interdendrite. This agrees with its large deviation of Al content and thus strength between dendrite and interdendrite.

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1. Introduction

Materials for high temperature applications, such as those in aerospace and petrochemical industries, are in high demand. Due to the incipient melting point at around 1300 °C, Ni-based superalloys have their applicability only up to temperatures between 1160 and 1277 °C [1,2]. Various alloys based on refractory metal elements having melting point higher than 1650 °C, such as niobium and tantalum alloys, might provide a practical solution to this problem [3–6]. It is because alloy softening does not generally occur until temperature exceeds 0.5–0.6 T_m [7], at which the material becomes susceptible to thermally activated deformation. Unfortunately, these two refractory alloy systems show high densities, poor oxidation resistance, and poor ductility at elevated temperatures. In considering the benefits of high-entropy and sluggish diffusion effects [8–12] which respectively enhance the formation of solid solution phase, and slow down the diffusion rate and thus creep rate, HEAs with suitable composition are considered as a good candidate to high-temperature applications. In recent years, Senkov et al. [13,14] proposed a refractory metal design concept to develop new alloys aimed at overcoming the temperature limitation of superalloys [15,16], which are named as refractory HEAs. The first two researches revealed that NbMoTaW and VNbMoTaW refractory HEAs possess a simple body-centered cubic (BCC) microstructure and pronounced solution hardening [13,14]. Both alloys still have

yield strengths of 405 and 477 MPa at 1600 °C, respectively. However, their drawbacks include much higher density than 9.0 g/cm³ and room-temperature brittleness. Later, they published the BCC alloy of HfNbTaTiZr [17,18] with slightly lower strength than the first two refractory HEAs but having excellent ductility exceeding 50%. However, this alloy still has a density higher than 9.0 g/cm³. Apparently, this alloy is unique in its large room-temperature ductility since BCC-HEAs are seldom ductile. In addition, this alloy also has quite high strength at elevated temperatures and thus is promising for high temperature applications. This finding has led researchers to investigate microstructure and properties of other refractory HEAs with various molar ratios of their constituent elements [10,19–24]. Among these, Senkov et al. reported four high-hardness and low-density refractory HEAs of the Cr–Nb–Ti–V–Zr system [22] and two Al-containing refractory HEAs: $\text{AlMo}_{0.5}\text{NbTa}_{0.5}\text{TiZr}$ and $\text{Al}_{0.4}\text{Hf}_{0.6}\text{NbTaTiZr}$ [24]. It was noted that the addition of Al could stabilize the bcc crystal structure and help to reduce the density of the refractory HEAs. The partial substitution of Hf with Al in the HfNbTaTiZr alloy was shown to reduce the density of the alloys by ~9%, increase the RT hardness by 29% and enhance yield strength by 98%. On the other hand, complete substitution of Cr with Al in the $\text{CrMo}_{0.5}\text{NbTa}_{0.5}\text{TiZr}$ alloy reduced the alloy density by 10.1%, increased RT hardness and yield strength by 12%, noticeably improved RT ductility and considerably increased, by more than 50%, high-temperature strength in the temperature range from 1073 K to 1473 K. Therefore, it is realized that Al addition could bring positive effects on the microstructure and properties of such refractory HEAs.

* Corresponding authors.

E-mail addresses: clin112@asu.edu, chunming@mx.nthu.edu.tw (C.-M. Lin), jwyeh@mx.nthu.edu.tw (J.-W. Yeh).

Based on the above consideration, this study selected Al for incorporation into HfNbTaTiZr for various kinds of benefits including lower density, and higher strength with adequate ductility. The evolution of microstructure, mechanical properties and fracture behavior of refractory $\text{Al}_x\text{HfNbTaTiZr}$ HEAs with the variation of x from 0 to 1 were investigated and discussed. Our aim has two more purposes: (1) to verify the ability of producing a single phase refractory solid solution with the addition of Al and (2) to optimize the Al content with best properties and provide a reference for further tuning the composition.

2. Experimental procedures

Bulk $\text{Al}_x\text{HfNbTaTiZr}$ refractory HEAs with various quantities of Al ($x = 0, 0.3, 0.5, 0.75, \text{ and } 1.0$), also simply denoted by Al_0 , $\text{Al}_{0.3}$, $\text{Al}_{0.5}$, $\text{Al}_{0.75}$, and $\text{Al}_{1.0}$, respectively which were prepared in a vacuum arc melting furnace under a high-purity argon atmosphere. Ingots were obtained in a copper mould. The purity of all raw elemental materials was better than 99.99 wt%. Each sample was reversed and re-melted four times in order to assure chemical homogeneity. The final samples were button-shaped, approximately 8 mm thick, with a shiny surface. The microstructure and chemical composition of the alloys were analyzed by scanning electron microscope (SEM, JEOL JEOL JSM-5410) and energy dispersive spectrometer (EDS). The crystal structures of the as-cast specimens were identified by X-ray diffraction (XRD, D/Max2500 V) using $\text{Cu K}\alpha$ radiation. Grain size was measured by the linear intercept method from micrographs taken by an optical microscope (OM, Olympus BX51M). Vickers hardness measurements were conducted using a MATSUZAWA SEIKI MV-1 hardness tester under a load of 10 kgf with the duration of 15 s. Seven measurements on each specimen were done to obtain an average. Room temperature compression tests were performed using an Instron 4468 machine under a strain rate of $10^{-3}/\text{s}$. In this test, we employed at least two cylindrical specimens 3 mm in diameter and 5 mm in length for each alloy, and put Teflon interface layers in-between specimen and platen for minimizing the friction and barrel effect.

3. Results and discussion

3.1. Crystal structure and density

X-ray diffraction patterns of HfNbTaTiZr, $\text{Al}_{0.3}\text{HfNbTaTiZr}$, $\text{Al}_{0.5}\text{HfNbTaTiZr}$, $\text{Al}_{0.75}\text{HfNbTaTiZr}$, and AlHfNbTaTiZr alloys are presented in Fig. 1. All peaks in the X-ray patterns are shown to be in agreement with a single BCC crystal structure with the exception that AlHfNbTaTiZr alloy has additional diffraction peaks. The exception was confirmed with three samples. It is thought to be originated from the interdendrite region which has relatively large Al segregation and a BCC structure with smaller lattice constant (see Section 3.3). All the peaks of BCC structure were also used to determine the lattice constants with a high degree of accuracy according to Bragg's law: HfNbTaTiZr ($a = 3.414 \pm 0.001 \text{ \AA}$), $\text{Al}_{0.3}\text{HfNbTaTiZr}$ ($a = 3.404 \pm 0.001 \text{ \AA}$), $\text{Al}_{0.5}\text{HfNbTaTiZr}$ ($a = 3.372 \pm 0.001 \text{ \AA}$), $\text{Al}_{0.75}\text{HfNbTaTiZr}$ ($a = 3.361 \pm 0.001 \text{ \AA}$ for Nb, Ta-rich dendrite and $a = 3.292 \pm 0.001 \text{ \AA}$ for Al, Hf, Zr-rich interdendrite, see Section 3.3), as shown in Table 1. Thus, the addition of Al reduces the lattice constants of the $\text{Al}_x\text{HfNbTaTiZr}$ alloys. The lattice constant of HfNbTaTiZr has been reported to be $a = 3.409 \text{ \AA}$ [17,18] and $a = 3.401 \text{ \AA}$ [19], respectively. Thus, the present lattice constant is slight higher than the reported data by a percentage within 0.4%.

The differences between the five alloys with regard to the relative intensities of these peaks are mainly due to the effects of cast texture under the excitation of a small number of grains within the illuminated volume by X-rays. All alloys except AlHfNbTaTiZr having two BCC structures display a single BCC structure. This indicates that the addition of Al does not change the BCC structure. This is not surprising when considering the tendency of Al to form strong BCC structure. It is well-known that although Al is an FCC element, it interacts with transition metal with hybrid pd -orbital and enhances the directional bonding for BCC structure [25–28]. The p -orbitals (filled) of Al as well as their high electron density and high Fermi level tend to promote the transfer of electrons to transition metals (TMs) with d -orbitals (partially filled), such as Ti, Zr, Nb and Hf. This results in the formation of covalent bonds in which Al–TM bonds are significantly shorter than the sum of the corresponding metallic radii. Because the TMs only have a partially filled d -shell, they contribute to only a general increase in covalency due to hybridization of the pd -orbitals, which leads to the formation of a BCC structure.

The formation of simple BCC solid solutions in these alloys again confirms the effect of high mixing entropy (ΔS_{mix}), which significantly reduces the Gibbs free energy of mixing (ΔG_{mix}) and provides greater stability in the formation of random solid solutions instead of intermetallic compounds or other ordered phases during solidification. The Hume-Rothery rules would predict large phase separation in these alloys even at high temperatures near melting range since their composing elements differ significantly in valence number (3 for Al, 4 for Hf, Ti and Zr, and 5 for Nb and Ta), in crystal structure (Al is FCC and Hf, Ti and Zr are HCP which transforms into BCC at high temperatures), and atomic size (the group of Al, Nb, Ta and Ti, and the group of Hf and Zr have atomic size differences around 9%) as seen in Table 1 [17,29–31]. Obviously, the Hume-Rothery rules with similar crystal structure, valence, and electronegativity, and small atomic size difference (<15%) for extended solubility can be relaxed to some extent due to the effects of high mixing entropy.

Using the rule-of-mixtures approach, theoretical lattice constants (a_{mix}) of an alloy can be estimated as follows:

$$a_{\text{mix}} = \sum c_i a_i$$

where c_i and a_i are atomic fraction and lattice constant of element i , respectively. Table 1 presents the lattice constants of elements, calculated results as well as the values obtained from X-ray diffraction. The lattice constants of elements except Al are from the research of Senkov et al. on similar alloy, in which the determination of lattice constants was fully discussed [17]. Because the lattice constant of BCC–Al cannot found in the literature, its lattice constant, $3.302 \text{ \AA}$, is calculated from its atomic radius, $1.430 \text{ \AA}$, in FCC–Al by using the closest atom packing approach. The theoretical lattice constants are calculated by substituting the actual overall compositions obtained by large-area EDS analyses (see Table 3) into the above equation. The standard deviation, ± 0.01 , has included the typical accuracy, ± 0.0005 atom fractions, of EDS analyses and the expanded accuracy range due to summation and multiplication in the equation. In addition, theoretical lattice constants are calculated by substituting the designed compositions. The standard deviation, ± 0.001 , is from the lattice constant data of each element [17]. It

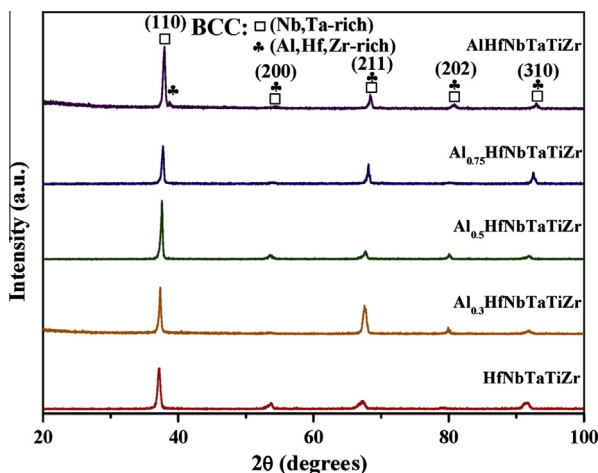


Fig. 1. X-ray diffraction patterns of $\text{Al}_x\text{HfNbTaTiZr}$ HEAs.

Table 1
Lattice constant (*a*), Vickers hardness (*Hv*), melting temperature (*T_m*), atomic radius (*r*), density (*ρ*), and crystal structure of Al, Hf, Nb, Ta, Ti, and Zr pure metals are listed in the upper part [17,29–31]. Theoretical lattice constant (*a_{mix}*) and experimental lattice constant (*a*), theoretical density (*ρ_{mix}*) and experimental density (*ρ*), theoretical hardness (*Hv_{mix}*) and experimental hardness (*Hv*) for the Al_xHfNbTaTiZr HEAs series are listed in the lower part.

Metal	Al		Hf	Nb	Ta	Ti	Zr	
<i>a</i> , Å	3.302		3.559	3.301	3.303	3.276	3.582	
<i>Hv</i> , kg/mm ²	167		180	135	89	99	92	
<i>T_m</i> , K	933		2506	2750	3290	1941	2128	
<i>r</i> , Å	1.430		1.541	1.429	1.430	1.418	1.551	
<i>ρ</i> , g/cm ³	2.70		13.31	8.57	16.69	4.51	6.52	
Crystal structure	Room temp.		FCC	HCP	BCC	BCC	HCP	
	High temp.		FCC	BCC	BCC	BCC	BCC	
Alloy	Theor. lattice constant (<i>a_{mix}</i> , Å)		Exper. lattice constant (<i>a</i> , Å)		Theor. density (<i>ρ_{mix}</i> , g/cm ³)		Exper. density (<i>ρ</i> , g/cm ³)	
					Theor. hardness (<i>Hv_{mix}</i>)		Exper. hardness (<i>Hv</i>)	
	Designed com.	Analyzed com.			Designed com.	Analyzed com.		
HfNbTaTiZr	3.404 ± 0.001	3.41 ± 0.01	3.414 ± 0.001		9.89 ± 0.01	9.96 ± 0.01	9.72 ± 0.04	119 ± 0.5
Al _{0.3} HfNbTaTiZr	3.398 ± 0.001	3.40 ± 0.01	3.404 ± 0.001		9.39 ± 0.01	9.48 ± 0.01	9.53 ± 0.02	113 ± 0.5
Al _{0.5} HfNbTaTiZr	3.395 ± 0.001	3.40 ± 0.01	3.391 ± 0.001		9.25 ± 0.01	9.25 ± 0.01	9.34 ± 0.01	109 ± 0.5
Al _{0.75} HfNbTaTiZr	3.391 ± 0.001	3.39 ± 0.01	3.372 ± 0.001		9.02 ± 0.01	9.10 ± 0.01	9.27 ± 0.01	106 ± 0.5
Al _{1.0} HfNbTaTiZr	3.387 ± 0.001	3.38 ± 0.01	3.361 ± 0.001 (Nb, Ta-rich)		8.56 ± 0.01	8.84 ± 0.01	8.91 ± 0.02	102 ± 0.5
			3.292 ± 0.001 (Al, Hf, Zr-rich)					

Table 2
Mixing enthalpies (kJ/mol) of unlike atomic pair [35,36].

ΔH_{mix}	Al	Hf	Nb	Ta	Ti	Zr
Al	0	−39	−18	−19	−30	−44
Hf	−39	0	4	3	0	0
Nb	−18	4	0	0	2	4
Ta	−19	3	0	0	1	3
Ti	−30	0	2	1	0	0
Zr	−44	0	4	3	0	0

can be seen that both theoretical lattice constants based on designed composition and analyzed composition have similar values and thus similar trend. Moreover, the differences between calculated and experimental lattice constants of the Al_xHfNbTaTiZr alloy are within 0.6% indicating the rule of mixtures holds well. This

is a good indication that the BCC structure is a fully disordered solid solution. In addition, the percentages of difference between calculated lattice constants based on analyzed compositions and experimental lattice constants for the Al₀, Al_{0.3}, Al_{0.5}, and Al_{0.75} are +0.12%, +0.12%, −0.26%, and −0.53%, respectively (not consider Al_{1.0} which has two BCC phases) indicating that higher Al content would result in even smaller lattice constant. Similarly, those between calculated lattice constants based on designed compositions and experimental lattice constants for the Al₀, Al_{0.3}, Al_{0.5}, and Al_{0.75} are +0.29%, +0.18%, −0.12%, and −0.56%, respectively. These two similar trends provide a good evidence of the effect of the strong bonding between Al and other constituent elements, as shown in Table 2 because the strong bonding could reduce the bond length and thus the lattice constant [32]. This finding is also consistent with the discussion on hybrid *pd*-orbital formation between Al and TMs mentioned above. The formations of such covalent bonds

Table 3
Designed and average actual compositions of each alloy and average compositions (at.%) in dendrite and interdendrite regions.

Alloy		Al	Hf	Nb	Ta	Ti	Zr
HfNbTaTiZr (Al ₀)	Designed composition	–	20.0	20.0	20.0	20.0	20.0
	Actual composition	–	20.8	18.9	20.2	20.2	19.9
	Interdendrite	–	23.4	16.5	13.9	20.6	25.6
	Dendrite	–	19.8	21.7	26.4	16.6	15.5
Al _{0.3} HfNbTaTiZr (Al _{0.3})	Designed composition	5.5	18.9	18.9	18.9	18.9	18.9
	Actual composition	6.6	18.7	18.6	18.4	18.4	19.3
	Interdendrite	5.8	20.9	16.1	12.8	20.2	24.2
	Dendrite	2.6	18.8	20.3	24.7	16.8	16.8
Al _{0.5} HfNbTaTiZr (Al _{0.5})	Designed composition	9.0	18.2	18.2	18.2	18.2	18.2
	Actual composition	9.4	18.0	17.6	17.6	18.3	19.1
	Interdendrite	12.7	21.1	13.5	11.3	16.8	24.6
	Dendrite	5.9	17.3	20.0	25.1	15.6	16.1
Al _{0.75} HfNbTaTiZr (Al _{0.75})	Designed composition	13.0	17.4	17.4	17.4	17.4	17.4
	Actual composition	12.5	16.9	17.4	17.6	17.6	18.0
	Interdendrite	15.2	19.6	11.7	9.7	17.1	26.7
	Dendrite	7.8	14.9	21.9	26.4	14.7	14.3
AlHfNbTaTiZr (Al ₁)	Designed composition	16.5	16.7	16.7	16.7	16.7	16.7
	Actual composition	16.2	16.4	16.9	16.4	17.4	16.6
	Interdendrite	19.6	19.8	12.0	9.0	14.8	24.8
	Dendrite	6.8	15.9	22.9	28.5	14.4	11.5

are significantly shorter than the sum of the corresponding metallic radii [25–28].

The density values of $\text{Al}_x\text{HfNbTaTiZr}$ alloys (both theoretical and experimental) are also presented in Table 1. Al addition indeed brings one important benefit in effectively reducing the actual density from 9.72 to 8.91 g/cm³. The theoretical densities (ρ_{mix}) of the alloy were calculated with the rule of mixtures:

$$\rho_{\text{mix}} = \frac{\sum c_i A_i}{\sum c_i A_i / \rho_i}$$

where A_i and ρ_i are the atomic weight and density of element i . For the theoretical density based on analyzed composition, they are calculated by substituting the actual overall compositions obtained by large-area EDS analyses (see Table 3). The standard deviation, ± 0.01 , has included the typical accuracy, ± 0.0005 atom fractions, of EDS analyses and the expanded accuracy range due to summation and multiplication in the equation. In addition, theoretical densities are calculated by substituting the designed compositions. The standard deviation, ± 0.01 , is from the hardness data of each element. It can be found that both theoretical data show similar trend as Al content increases. Besides this, the higher measured densities of Al-containing alloys and the increased density difference with increasing Al content also confirm the effect of strong bonding between Al and other constituent elements.

3.2. Hardness analysis

The Vickers hardness values (H_v) of $\text{Al}_x\text{HfNbTaTiZr}$ HEAs are tabulated in Table 1 and plotted as Fig. 2. The addition of Al increases the hardness of $\text{Al}_x\text{HfNbTaTiZr}$ alloys. This could be firstly explained by the difference in atomic size, which can induce significant localized lattice distortion and has a contribution to solution hardening. From Table 1, Al has an atomic radius similar to those of Nb, Ti, and Ta, but distinctly smaller than that of Hf and Zr. According to the following equation, which is commonly used for the calculation of overall differences in atomic size, δ [33,34], the δ with an amplifying factor 100 for Al_0 to $\text{Al}_{1.0}$ alloys are 3.03, 3.09, 3.12, 3.15 and 3.17, respectively. These small values indicate the present BCC phase is essentially disordered solid solution since Guo et al. found that disordered solid solutions should form when $\delta < 0.066$ [33,34]. Furthermore, the increase of δ agrees with the increasing trend of hardness.

$$\delta = 100 \sqrt{\sum_{i=1}^n c_i \left(1 - \frac{r_i}{\bar{r}}\right)^2}$$

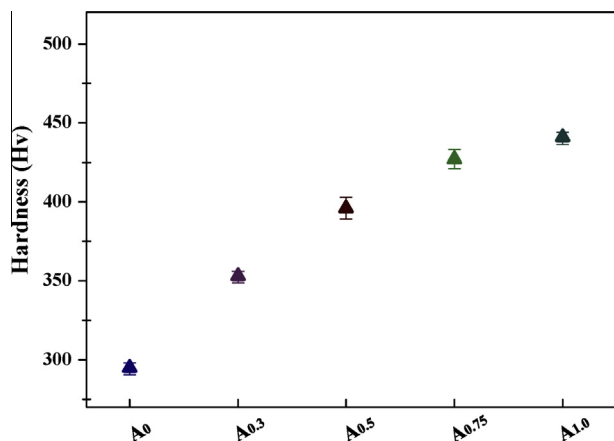


Fig. 2. Hardness values of $\text{Al}_x\text{HfNbTaTiZr}$ HEAs.

where $\bar{r} = \sum_{i=1}^n c_i r_i$, and c_i and r_i are the atomic fraction and atomic radius of element i .

In addition, the distinctly strong bonds between Al and the constituent transition metals, as shown in Table 2, also have a strong hardening effect. To reveal this factor, the rule-of-mixtures approach is used to predict the theoretical hardness values ($H_{v_{\text{mix}}}$) of an alloy with the following equation:

$$H_{v_{\text{mix}}} = \sum c_i (H_v)_i$$

where c_i and H_{v_i} are atomic fraction and hardness of element i . Both the theoretical hardness values based on analyzed compositions and based on designed compositions are calculated, respectively as listed in Table 1. By comparison, the two theoretical hardness values for each alloy are very close to each other. It is noted from Table 1 that the experimental hardness values are much higher than the theoretical values. Moreover, the increase in hardness ranges from 2.5 to 4.3 folds and higher Al content gives larger folds. Senkov et al. have reported similar instances of solution hardening in other refractory HEAs [24]. The partial substitution of Hf with Al from HfNbTaTiZr to $\text{Al}_{0.4}\text{Hf}_{0.6}\text{NbTaTiZr}$ alloy was shown to increase the RT hardness by 29% and a complete substitution of Cr with Al in the $\text{AlMo}_{0.5}\text{NbTa}_{0.5}\text{TiZr}$ alloy increased RT hardness and yield strength by 12%. This is attributed to the strong bonding of Al and other transition elements in addition to the atomic-size-difference effect.

3.3. Microstructure and composition analysis

As-cast specimens of $\text{Al}_x\text{HfNbTaTiZr}$ alloys have been observed under optical microscope to evaluate the grain size by linear intercept method. The Al_0 , $\text{Al}_{0.3}$, $\text{Al}_{0.5}$, $\text{Al}_{0.75}$, and $\text{Al}_{1.0}$ specimens have grain sizes of $120 \pm 23 \mu\text{m}$, $105 \pm 12 \mu\text{m}$, $86 \pm 9 \mu\text{m}$, $89 \pm 13 \mu\text{m}$, $92 \pm 13 \mu\text{m}$, respectively. This indicates that an increase in Al content reduces grain size at first but has no obvious reduction for $\text{Al}_{0.5}$, $\text{Al}_{0.75}$, and $\text{Al}_{1.0}$ alloys.

Fig. 3 presents backscattered electron images (BEI) of the $\text{Al}_x\text{HfNbTaTiZr}$ as-cast specimens, which show that each microstructure contains dendrite and interdendritic phase regions. Table 3 lists the actual overall compositions of the various specimens, as determined by large-area EDS analysis. It shows that the actual compositions are very close to the designed compositions of $\text{Al}_x\text{HfNbTaTiZr}$ alloys, respectively. At $x = 0$, the bright dendrite region is enriched with Nb and Ta, while the dark and gray interdendrite region is enriched with Hf, Ti, and Zr, as shown in Fig. 3(a). This segregation trend is similar to that analyzed for the same alloy [17,19]. At $x = 0.3$ – 1.0 , Al segregates to interdendrite region and intensifies with increased Al addition. In contrast, Ta and Nb concentrations in dendrite region are increased relative to interdendrite region correspondingly. It is interesting to note that Ti shows less segregation between dendrite and interdendrite region as compared with other elements and becomes lesser segregation with increased Al addition. The observed partitioning of elements in the present alloys could be attributed to two main factors: differences in the melting point (see Table 1) and the driving of bonding energy (see Table 2). Ta, Nb-rich dendrite first forms simply because these two elements were the highest two in melting points. This was also used to explain the segregation of highest-melting-point elements in the dendrite of refractory alloys [15,19]. According to phase diagram, dendrite forms from liquid phase to solid–liquid phase field. Subsequent, solidification would complete from solid–liquid region into solid phase region. Therefore, the solid–liquid region of the present alloys would be between Nb, Ta-rich solid phase and Nb, Ta-depleted liquid phase. On the other hand, Al has the lowest melting point and stronger bonding with Hf, Ti, and Zr, both factors enhance Al segregation

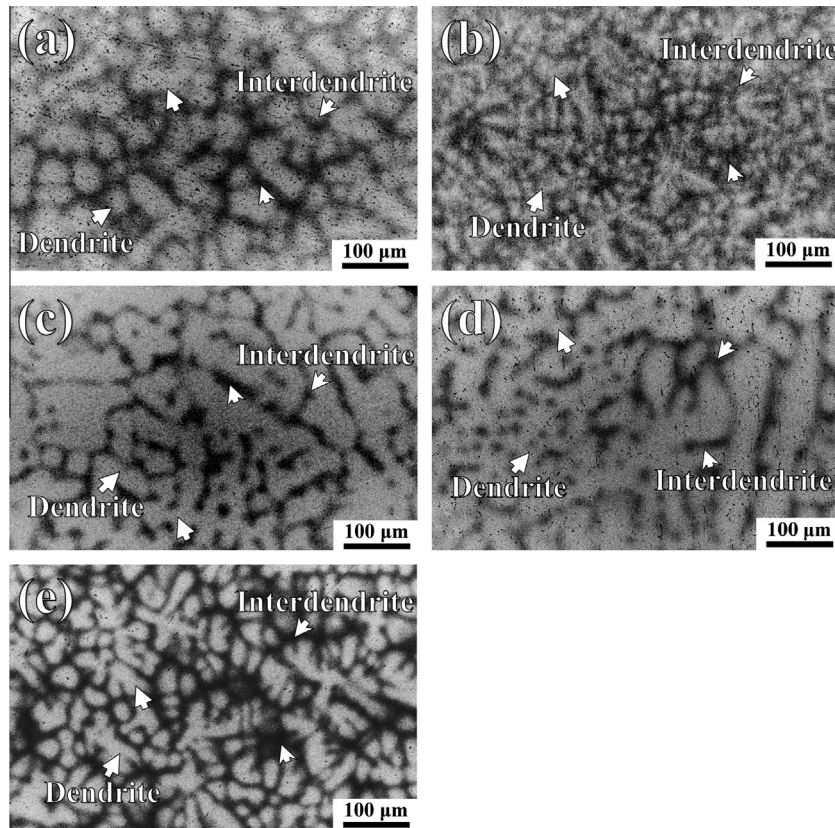


Fig. 3. BEI images of $\text{Al}_x\text{HfNbTaTiZr}$ HEAs: (a) HfNbTaTiZr , (b) $\text{Al}_{0.3}\text{HfNbTaTiZr}$, (c) $\text{Al}_{0.5}\text{HfNbTaTiZr}$, (d) $\text{Al}_{0.75}\text{HfNbTaTiZr}$, and (e) AlHfNbTaTiZr alloys.

into interdendrite. This suggests that solvus boundary of the hyperspace phase diagram encompasses from Nb, Ta-rich solid at higher temperatures to Al, Hf, Ti-rich solid at lower temperatures. However, the bonding between Al and Ti is less than that of Al–Hf and Al–Zr. As a result, Ti shows lesser segregation as Al content increases.

The extra diffraction peaks existing in XRD pattern of AlHfNbTaTiZr could be explained now. As aluminum content in the dendrite of this alloy is 6.8 at.% and that in the interdendrite is 19.6 at.% (see Table 3), their theoretical lattice constants based on the rule of mixtures can be calculated to be $3.378 \pm 0.002 \text{ \AA}$ and $3.418 \pm 0.001 \text{ \AA}$, respectively. The deviation between these two is 1.2%. However, experimental lattice constants of dendrite and interdendrite are $3.361 \pm 0.001 \text{ \AA}$ and $3.292 \pm 0.001 \text{ \AA}$, respectively and have a reversed mutual deviation of 2.1%. This deviation becomes large enough to be resolved in XRD pattern. Furthermore, this reversed deviation again confirms the strong bonding effect between Al and other constituent elements since higher Al content significantly enhances the reduction of lattice constant. This is reasonable because stronger bond strength could result in shorter bond length and thus smaller lattice constant [25–28,32].

3.4. Compressive properties

3.4.1. Stress–strain curves

Fig. 4 presents the typical stress–strain curves for $\text{Al}_x\text{HfNbTaTiZr}$ HEAs measured at room temperature. 0.2% offset yield strength, apparent Young's modulus (containing machine modulus effect), and fracture strain are summarized in Table 4. It demonstrates that yield strength increases remarkably with increased Al content. AlHfNbTaTiZr alloy attains the yield strength of $1486 \pm 7 \text{ MPa}$. This is consistent with hardness trend and also

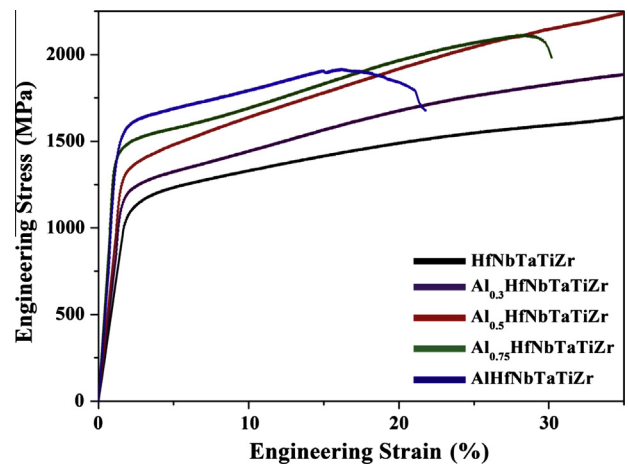


Fig. 4. Compressive engineering stress–strain curves of $\text{Al}_x\text{HfNbTaTiZr}$ HEAs.

attributable to solution hardening effect arising from atomic size difference and strong bonding between Al and other constituent elements mentioned above. However, ductility shows a gradual decrease, but AlHfNbTaTiZr still has a ductility of 21%. The apparent Young's modulus increases significantly with increased Al content and reaches $103 \pm 1 \text{ GPa}$ for AlHfNbTaTiZr alloy. The higher apparent Young's modulus is in agreement with the increased strong bonding between Al and other composing metals since atomic size difference is thought to have no relation with Young's modulus.

It is noted that the $\text{Al}_{0.4}\text{Hf}_{0.6}\text{NbTaTiZr}$ alloy reported by Senkov et al. [24] has much higher strength, 1841 MPa, and much lower fracture strain, 10%, than $\text{Al}_{0.4}\text{HfNbTaTiZr}$ alloy. The strength and

Table 4

Compressive yield strength, apparent Young's modulus, and fracture strain of $\text{Al}_x\text{HfNbTaTiZr}$ HEAs.

	Yield strength (MPa)	Young's modulus E (GPa)	Fracture strain (%)
HfNbTaTiZr	1073 ± 6	55 ± 1	>50
$\text{Al}_{0.3}\text{HfNbTaTiZr}$	1188 ± 7	63 ± 3	>50
$\text{Al}_{0.5}\text{HfNbTaTiZr}$	1302 ± 6	97 ± 2	46
$\text{Al}_{0.75}\text{HfNbTaTiZr}$	1415 ± 4	102 ± 2	30
AlHfNbTaTiZr	1489 ± 7	103 ± 1	21

fracture strain of the latter alloy are 1247 MPa, and >46%, respectively by taking the average from $\text{Al}_{0.3}\text{HfNbTaTiZr}$ and $\text{Al}_{0.5}\text{HfNbTaTiZr}$ alloys. The discrepancy could be attributable to their different compositions, and processing. The difference in composition is better revealed by atomic percentage: $\text{Al}_{7.5}\text{Hf}_{18.5}\text{Nb}_{18.5}\text{Ta}_{18.5}\text{Ti}_{18.5}\text{Zr}_{18.5}$ versus $\text{Al}_8\text{Hf}_{12}\text{Nb}_{20}\text{Ta}_{20}\text{Ti}_{20}\text{Zr}_{20}$. Each component differs in percentage in these two alloys. Besides, the present alloys were investigated in the as-cast state whereas $\text{Al}_{0.4}\text{Hf}_{0.6}\text{NbTaTiZr}$ alloy was HIPed at 1473 K and 207 MPa for 2 h, then annealed at 1473 K for 24 h in continuously flowing high-purity argon, and finally cooled at a rate of 10 K min^{-1} . Because the difference in total mixing enthalpy is small between $\text{Al}_{0.4}\text{Hf}_{0.6}\text{NbTaTiZr}$ and $\text{Al}_{0.4}\text{HfNbTaTiZr}$ alloy based on Table 2 and cannot account for the discrepancy of strength, homogenization to maximize solution hardening and/or slow cooling to induce precipitation hardening might be the possible factors. However, it needs further study to clarify the mechanisms.

In the single BCC structure of $\text{Al}_x\text{HfNbTaTiZr}$ HEAs, each atom can be regarded as a solute atom and thus the matrix is whole-solute. Nevertheless, variations in the size and properties of these solute atoms enable them to elastically distort the crystal lattice, introducing local elastic stress fields. Since the interactions among these local elastic stress fields and the stress field of dislocations can hinder dislocation movements, it can be expected that this whole-solute matrix would give a large solution hardening effect as compared with conventional matrix based on one major element. In conventional alloy, the relationship between strengthening ($\Delta\sigma$) and solute concentration (c) is generally expressed as [37,38]:

$$\Delta\sigma \propto c^n$$

where $n \approx 0.5$. In present HEAs, Al concentration in the alloy can be calculated by $c = x/(5 + x)$ if we treat $\text{Al}_x\text{HfNbTaTiZr}$ HEAs as

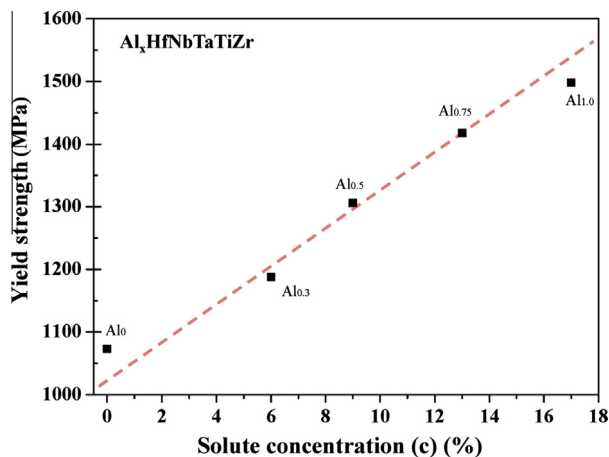


Fig. 5. Effects of solute concentration of Al on the yield strength of the $\text{Al}_x\text{HfNbTaTiZr}$ HEAs.

quasi-binary alloy, i.e., $\text{Al}_x-(\text{HfNbTaTiZr})$. Fig. 5 shows yield strength as a function of c . It can be seen that the distribution of data fits a linear line very well and can be expressed by the following equation:

$$\sigma = 1031 + 26.1c \text{ (MPa)}$$

This shows that yield strength increases linearly with the concentration of Al in the alloy. Because the grain sizes from microstructure observation are in the same order as observed in Section 3.3, grain size strengthening effect by Hall–Petch equation could be neglected. Therefore, this linear relationship suggests that if we want to see the strengthening effect of a certain element in a single-phase HEA alloy, quasi-binary alloy concept might be used. The effect is linear by nature as long as no second phase forms. Besides this, it is noted that the partitioning of Al in the as-cast structure has no obvious effect on the linear relationship. This further suggests that the lattice distortion, Young's modulus, strength, and strengthening of dendritic solid solution and interdendritic one could be averaged as if the composition is uniform throughout the material. The fact of linearity is thus originated from the strengthening contribution per Al atom. This proportionality especially reflects the randomness of Al atoms in the solid solution, which is consistent with the conclusion drawn from the discussion on lattice constant and density in Section 3.1.

Senkov et al. made theoretical derivations based on the atomic size misfit and modulus misfit of atoms in solutes and solvents in order to explain the high degree of solution hardening observed in HfNbTaTiZr [17]. Treating this material as a quasi-binary alloy in which Ta, Nb, and Ti were solvents and 40 at.% of Hf plus Zr were solutes led to a predicted yield strength of 1094 MPa, which was in reasonable agreement with the observed value of 929 MPa. It is interesting to note that their prediction of yield strength of HfNbTaTiZr is very close to the 1073 ± 6 MPa obtained in this study (see Table 4). This means that their derivation and prediction are accurate for this alloy if the present data is considered. Obviously, their model for the strengthening reason of one composition, HfNbTaTiZr, cannot be applied to the present alloy series since Al having distinctly strong bonding with other constituent elements is varied in content. In the present alloy series, strong bonding factor becomes a new contributor to the strengthening, which in combination with atomic-size-difference factor (or lattice distortion), follows linearity with Al content.

3.4.2. Fracture behavior

Fig. 6 presents longitudinal-section microstructure of $\text{Al}_x\text{HfNbTaTiZr}$ specimens after compression test. The microstructure of the Al_0 specimen (see Fig. 6(a)) deformed under 50% compressive strain does not fracture but shows shear band with severe grain flow along it, as indicated by arrows. Similar morphologies were also observed in the $\text{Al}_{0.3}$ alloy. Fig. 6(b) is that of $\text{Al}_{0.5}$ specimen deformed and fractured at 46% strain, showing some cracks and the loss of a portion departed from the fracture specimen. Severe grain flow along shear band still formed. This indicates large plastic deformation of microstructure and is in agreement with its excellent ductility. Fig. 6(c) presents $\text{Al}_{0.75}$ specimen deformed at 30% strain. It shows 45° shear band and crack along it. However, moderate grain flow and dimple rupture mode can be seen in consistency with its moderate ductility. In contrast, the SEI image of $\text{Al}_{1.0}$ specimen deformed at fracture strain of 21% in Fig. 6(d) shows many off- 45° cracks and small voids. Close investigation demonstrates that crack initiations are mainly along the boundaries between dendrite and interdendrite. This evidence reveals that there is a large deformation difference and thus stress concentration between dendrite and interdendrite. This is consistent with the large deviation of Al content and strength between dendrite and interdendrite, which causes a large stress concentration at

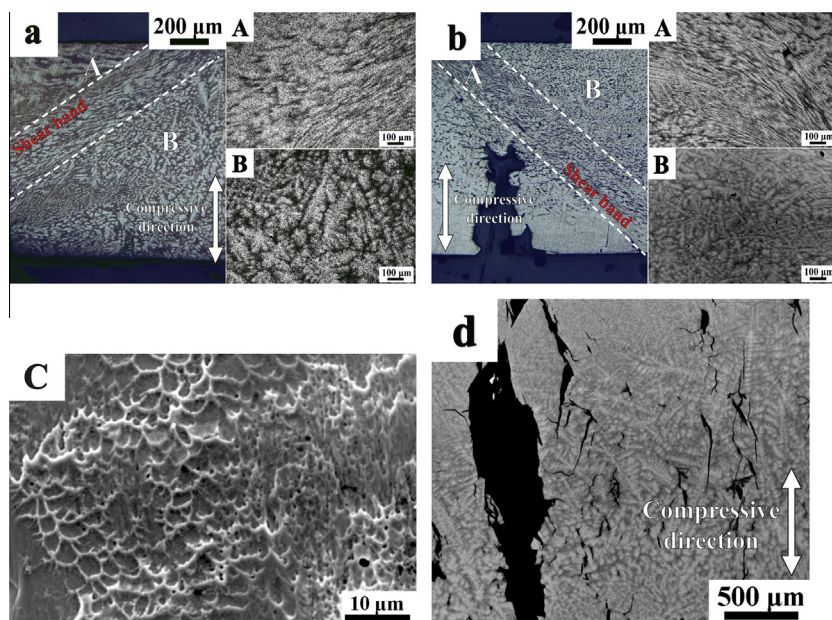


Fig. 6. Longitudinal-section showing the microstructure of deformed $\text{Al}_x\text{HfNbTaTiZr}$ specimens: (a) OM image of Al_0 specimen deformed under 50% compressive strain with SEI images: A-area remot from shear band and B-area near shear band; (b) OM image of $\text{Al}_{0.5}$ specimen deformed under 46% compressive strain with SEI images: A-area remot from shear band and B-area near shear band; (c) OM image of $\text{Al}_{0.75}$ specimen deformed under 30% compressive strain with SEI images showing dimple-rupture; (d) SEI image of $\text{Al}_{1.0}$ specimen deformed under 21% compressive strain.

the interface regions between soft dendrite and hard interdendrite. This suggests that homogenization might improve the concentration difference, stress concentration, and thus the overall ductility. 4. Conclusions Al has been selected to improve the strength and reduce the density of very ductile BCC HfNbTaTiZr alloy. The resulting $\text{Al}_x\text{HfNbTaTiZr}$ alloys ($x = 0, 0.3, 0.5, 0.75$, and 1.0) maintain a single BCC solid solution with typical dendritic structures in its as-cast state. The addition of Al increased the compressive yield strength from 1073 ± 6 MPa to 1489 ± 7 MPa; however, ductility was reduced from >50 to 21%. The increase in yield strength can be attributed to solid solution strengthening resulting from increased lattice distortion and stronger bonding between Al and constituent transition metals. The solution hardening shows a linear relation between yield strength and atomic percentage of Al, which suggests that the explanation of strengthening effect of a certain element in a single-phase HEA alloy can use quasi-binary alloy concept and the effect is linear as long as no second phase forms. The decrease in ductility can be attributed to the effect of increased strength which hinders plastic deformation. Crack formations in AlHfNbTaTiZr alloy with the lowest fracture strain are mainly along the boundaries between dendrite and interdendrite. This indicates that there is large deformation difference and thus stress concentration between dendrite and interdendrite, agreeing with the large deviation of Al content and thus strength between dendrite and interdendrite.

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